

SIMATS SCHOOL OF ENGINEERING

ASSESSMENT TOOL – 1

Course Name: Internet Programming

Course Code: CSA4305

Student Name: Mathusri P

Register Number: 192311363

Question 1: Online Symposium Portal

Given HTML Code

```
<form action="/register" method="POST">
    <input type="text" name="name">
    <input type="email" name="email">
    <button>Submit</button>
</form>
```

1. Identify appropriate HTML5 semantic elements missing in the structure.

The following semantic elements are missing:

- <header> – Displays page heading.
 - <main> – Contains the main content.
 - <section> – Groups the registration form.
 - <form> should include <label> elements for inputs.
 - <footer> – Displays copyright or contact details.
-

2. Modify the structure to make it standards-compliant.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Online Symposium Registration</title>
</head>
<body>

<header>
```

```
<h1>Online Symposium Registration</h1>
</header>

<main>
  <section>
    <form action="/register" method="POST">

      <label for="name">Name:</label>
      <input type="text" id="name" name="name" required>

      <br><br>

      <label for="email">Email:</label>
      <input type="email" id="email" name="email" required>

      <br><br>

      <button type="submit">Submit</button>

    </form>
  </section>
</main>

<footer>
  <p>&copy; 2026 SIMATS</p>
</footer>

</body>
</html>
```

3. Interpret the HTTP request generated during form submission.

- The browser sends an **HTTP POST** request to the server.
- The URL is **/register**.
- The entered name and email are sent in the request body.
- The server receives the submitted data for processing.

Example:

POST /register HTTP/1.1

Host: example.com

Content-Type: application/x-www-form-urlencoded

name=Mathusri&email=mathusri@gmail.com

4. Analyze the HTTP response cycle from server to browser.

1. User fills the form.
 2. Browser sends HTTP POST request.
 3. Server validates the data.
 4. Server stores the information.
 5. Server sends an HTTP response (200 OK or redirect).
 6. Browser displays the confirmation page.
-

5. Suggest improvements for usability and accessibility.

- Add <label> for every input field.
- Use the required attribute.
- Display clear error messages.
- Use meaningful button text.
- Make the page mobile responsive.
- Use proper heading hierarchy.

Question 2: Cross-Browser Compatibility Issue

Given HTML Code

```
<html>

<head>

<title>Company</title>

</head>


<body>


<div>

<h1>Welcome

<p>About us

</div>


</body>

</html>
```

1. Identify improper nesting and missing closing tags.

Issues:

- Missing <!DOCTYPE html>
- Missing lang attribute.
- <h1> tag is not closed.
- <p> tag is not closed.
- Improper nesting of HTML elements.
- Missing semantic HTML elements.

2. Analyze how different browsers may interpret this structure.

- Browsers automatically close missing tags differently.
 - Chrome may display the page correctly.
 - Firefox and Edge may render the layout differently.
 - This causes inconsistent appearance across browsers.
-

3. Identify missing semantic elements.

Missing semantic tags include:

- <header>
 - <main>
 - <section>
 - <footer>
-

4. Provide a corrected W3C-compliant HTML structure.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Company</title>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
  <h1>Welcome</h1>
```

```
</header>
```

```
<main>
```

```
  <section>
```

```
    <p>About us</p>
```

```
  </section>
```

</main>

<footer>

<p>© 2026 Company</p>

</footer>

</body>

</html>

5. Suggest techniques to ensure cross-browser compatibility.

- Follow W3C HTML standards.
- Validate HTML using the W3C Validator.
- Use CSS Reset or Normalize.css.
- Test on Chrome, Firefox, Edge, and Safari.
- Use responsive web design.
- Avoid deprecated HTML tags.

Question 3: Form Submission Failure

Given HTML Code

```
<form action="/submit" method="GET">
```

```
<input type="text" name="username">
```

```
<input type="password" name="password">
```

```
<button type="button">Login</button>
```

```
</form>
```

1. Identify issues in the form configuration.

Problems:

- Button type is "button" instead of "submit".
 - Password should not be sent using GET.
 - Missing labels.
 - Missing required attribute.
-

2. Analyze why the HTTP request is not triggered.

- A button with type="button" performs no action by default.
 - Since it is not a submit button, the browser never submits the form.
 - Therefore, no HTTP request is sent to the server.
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3. Interpret the role of GET method in this scenario.

- GET sends data through the URL.
- Username and password become visible in the browser address bar.
- GET is suitable for searching data.
- It is **not secure** for login forms.

Example:

/submit?username=Mathusri&password=12345

4. Propose a corrected implementation.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Login</title>
```

```
</head>
```

```
<body>
```

```
<form action="/submit" method="POST">
```

```
<label for="username">Username:</label>
```

```
<input type="text" id="username" name="username" required>
```

```
<br><br>
```

```
<label for="password">Password:</label>
```

```
<input type="password" id="password" name="password" required>
```

```
<br><br>
```

```
<button type="submit">Login</button>
```

```
</form>
```

```
</body>
```

```
</html>
```

5. Suggest improvements for secure data transmission.

- Use **POST** instead of GET.
- Enable **HTTPS** for encrypted communication.
- Validate input on both client and server.
- Store passwords securely using hashing.
- Implement authentication and session management.
- Protect against SQL Injection and Cross-Site Scripting (XSS).

Code of Conduct

I, Mathusri P-192311363, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mathusri P

